`dr-iissi` Theme Guide

Usage reference and visual regression deck

IISSI Marp theme

UNIVERSIDAD D SEVILLA

What this deck is for

Learn how to use the `dr-iissi` theme in real slides.

See the supported layout patterns and content conventions.

Verify typography, code blocks, math, image framing, footer, and pagination.

Use the later sections as a visual regression deck after theme or engine changes.

Quick start

Use the theme with Marp CLI:

```
marp --theme-set ./css/dr-iissi.css --engine ./scripts/marp-engine.js slide.md --pdf
```

Recommended front matter in a deck that lives outside `marp-theme/`:

```
---
marp: true
theme: dr-iissi
marp-theme-rel-dir: ../../marp-theme
paginate: true
size: 16:9
---
```

Theme-specific front matter

`marp-theme-rel-dir` tells the engine where theme assets live relative to the current markdown file.

- In this guide the value is `..`
- In `iissi-1/t*/...md` files the value is `../../marp-theme`
- Use a relative path so PDF builds stay portable across machines

If the path is wrong, background images and local fonts may not resolve during render.

Column marker rules

For `two-col`, `two-col-3-7`, `two-col-7-3`, and `three-col`:

1. Use `*** left ***` to start the left column.
2. Use `*** right ***` to start the right column.
3. Use `*** center ***` only in `three-col`.
4. Do not use `<hr/>` inside column layouts.

Optional: add `v-align` to vertically center the slide body.

Guide map

Section 1. Typography and code fonts

Section 2. Base layouts and images

Section 3. Vertical alignment

Section 4. `mr`, `sql`, and KaTeX

Section 5. Visual regression cases A-J

Section 6. Realistic topic examples T1-T12

Section 7. Mixed regression cases K-N

Theme font: default (`Roboto`)

This slide uses the default theme font.

- Accents and symbols: áéíóú ñ ç
- **Bold sample**
- *Italic sample*
- `Inline code sample`

Theme font override: `Lato`

Use slide classes such as `font-lato` when a deck needs a different tone without changing the whole theme.

- Paragraph rhythm sample
- **Bold sample**
- *Italic sample*
- `Inline code sample`

Theme font override: `Roboto Condensed`

This variant is useful for denser titles or compact text blocks.

- Narrower letterforms
- Good for labels and summaries
- `Inline code sample`

Theme font override: `Source Sans 3`

Use this slide to validate the local variable font and italic face.

- Normal text sample
- *Italic sample*
- **Bold sample**

Theme font override: `Caveat`

This slide is intentionally distinctive so font loading issues are obvious.

- Handwritten tone sample
- Long enough sentence to inspect rhythm and contrast.
- **Bold sample**
- *Italic sample*

Code font: default stack

Default code blocks use the theme monospace stack.

```
SELECT
  d.nombre AS departamento,
  e.id_emp,
  e.nombre,
  e.salario,
  ROW_NUMBER() OVER (
    PARTITION BY d.id_dept
    ORDER BY e.salario DESC
  ) AS orden_salario
FROM Empleado e
JOIN Departamento d ON d.id_dept = e.id_dept
WHERE e.salario IS NOT NULL
ORDER BY d.nombre, orden_salario;
```

Code font: `JetBrains Mono`

The engine supports a local JetBrains Mono variant through the fence info string.

```
Empleado(PK id_emp, nombre, salario, FK id_dept null)
Departamento(PK id_dept, nombre AK, presupuesto)
Proyecto(PK id_proy, nombre, FK id_dept)
Asignacion(PK id_emp, PK id_proy, horas, fecha_inicio)
```

```sql
SELECT e.nombre, p.nombre, a.horas
FROM Asignacion a
JOIN Empleado e ON e.id_emp = a.id_emp
JOIN Proyecto p ON p.id_proy = a.id_proy
WHERE a.horas ≥ 20
ORDER BY a.horas DESC, e.nombre;
```

Code fence selectors

Pick the code font in the fence info string:

- `default` or `consolas` for the default monospace stack
- `jetbrains` or `jb` for `JetBrains Mono`

Examples:

```
```sql default
SELECT * FROM Empleado;
```
```

```
```mr jetbrains
Empleado(PK id_emp, nombre)
```
```

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Layout: `default`

Standard single-column content slide.

- Main point one
- Main point two
- Subpoints can be nested
- Suitable for most explanatory slides

Headings: `h2`, `h3`, `h4`

This is an `h2`

This is an `h3`

This is an `h4`

Body text should remain readable below centered secondary headings.

Layout: `content`

> Use this style for agendas and section summaries

> The layout is intentionally lighter than `default`

> It works well with short, scannable bullet lists

Layout: `default` with a full image

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Layout: `default` with text and image

- Text may appear before a framed image.
- Images should keep their box styling.
- Spacing should remain consistent.

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Layout: `two-col`

Left column

Content for the first half.

- Item A
- Item B
- Item C

Right column

Content for the second half.

- Item D
- Item E
- Item F

Layout: `three-col`

First column

Text for column one.

Second column

Text for column two.

Third column

Text for column three.

Layout: text + code + image

Quick summary:

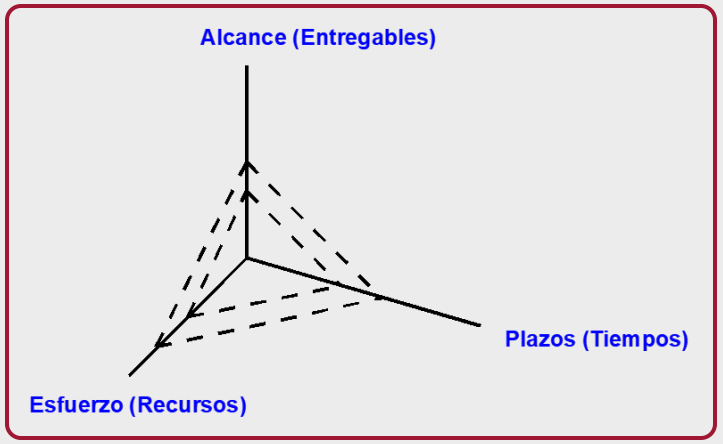
- Context
- Rule
- Conclusion

```
SELECT nombre, salario
FROM Empleados
WHERE salario > 2000;
```

Layout: `three-col v-align`

Vertically centered text.

```
Pedido(PK id_ped, FK id_cli)
Cliente(PK id_cli, nombre)
```



Layout: `two-col-3-7`

Left column at 30%.

Short contextual content.

The right column gets the main space.

- Detailed point one
- Detailed point two
- Detailed point three

Layout: `two-col-7-3`

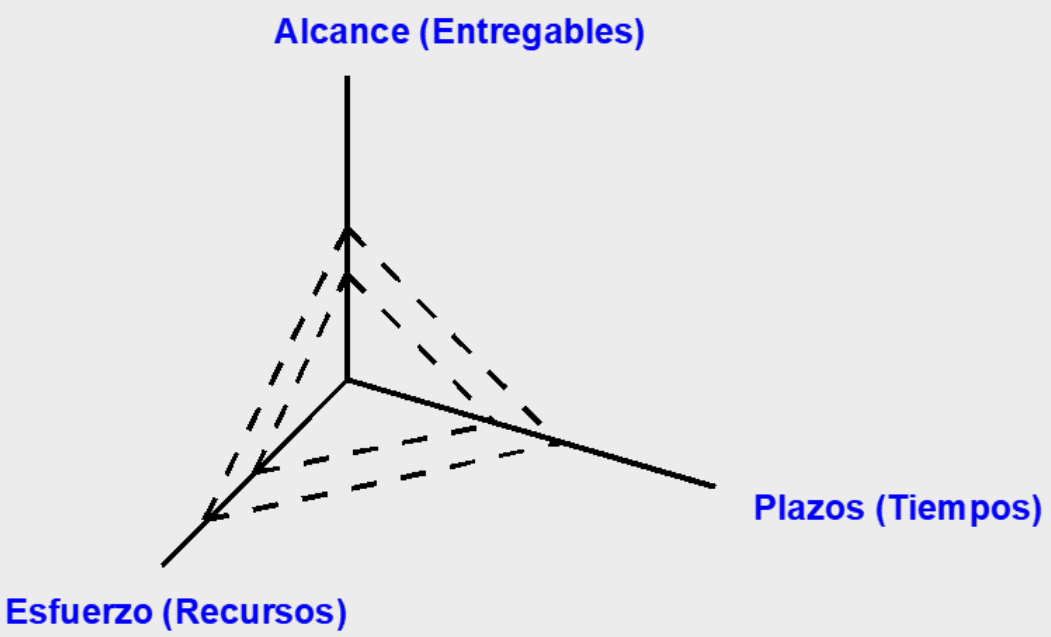
The left column holds the main explanation.

- Main concept
- Supporting point
- Partial conclusion

The right column is suitable for notes or compact media.

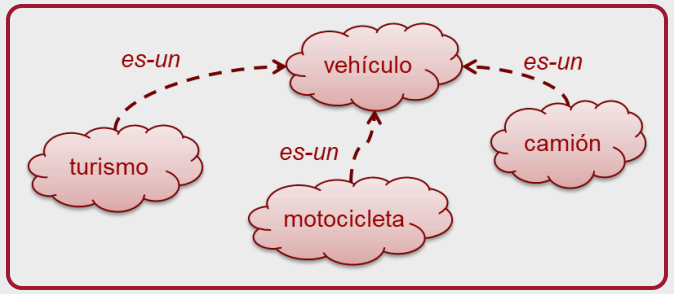
Layout: image and text in 50/50

Text above an image.



- Feature A
- Feature B

Layout: image-heavy `two-col-3-7`



Caption-like text below the image.

- More detailed explanation
- Supporting note
- Closing point

Layout: text opposite image

Text at the left side of the image.

- Feature A
- Feature B

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Layout: text-heavy `two-col-7-3`

Primary explanation with more horizontal room.

- Point one
- Point two
- Point three

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`v-align` on `default`

This block should appear vertically centered inside the slide body.

`v-align` on `content`

Item 1

Item 2

Item 3

`v-align` on `two-col`

Text in the left column.

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`v-align` on `two-col-3-7`

Short text.

```
SELECT *
FROM Empleados
WHERE salario > 2000;
```

`v-align` on `two-col-7-3`

```
Empleado(PK id_emp, nombre, FK id_dept)
Departamento(PK id_dept, nombre)
```

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`mr` code blocks

Minimal relational-model sample used to validate custom highlighting.

```
Empleado(PK id_emp, nombre, apellidos, salario, FK id_dept null)
Departamento(PK id_dept, nombre AK, presupuesto)
Proyecto(PK id_proy, nombre, FK id_dept)
Asignacion(PK id_emp, PK id_proy, horas, fecha_inicio)
```

`sql` code blocks

Longer SQL sample used to inspect indentation, wrapping, and readability.

```sql
SELECT
  e.id_emp,
  e.nombre,
  d.nombre AS departamento,
  e.salario
FROM Empleado e
JOIN Departamento d ON e.id_dept = d.id_dept
WHERE e.salario > 30000
  AND d.presupuesto >= 100000
ORDER BY d.nombre, e.salario DESC, e.nombre;
```

`python`, `javascript`, `html`, and `css` code blocks

Same code-block treatment as `mr` and `sql`, with distinct background colors.

```python
def salario_neto(bruto, retencion=0.15):
  return round(bruto * (1 - retencion), 2)
```

```javascript
const empleados = datos.filter((e) => e.salario > 30000);
console.log(empleados.map((e) => e.nombre));
```

```html
<section class="empleados">
  <h2>Departamento</h2>
  <p>Listado actualizado</p>
</section>
```

```css
.empleados {
  border: 2px solid #9f1732;
  padding: 0.75rem 1rem;
  background: #fff;
}
```

`mr` + `sql` in columns

```
Empleado(PK id_emp, nombre, salario,
Departamento(PK id_dept, nombre AK, p
Asignacion(PK id_emp, PK id_proy, hor
```

```sql
SELECT
  e.nombre,
  d.nombre AS departamento,
  SUM(a.horas) AS total_horas
FROM Empleado e
JOIN Departamento d ON d.id_dept = e.id_dept
LEFT JOIN Asignacion a ON a.id_emp = e.id_emp
WHERE e.id_dept IS NOT NULL
GROUP BY e.nombre, d.nombre
HAVING SUM(a.horas) > 10
ORDER BY total_horas DESC, e.nombre;
```

KaTeX formulas

$$\prod_{\text{columns}} \left(\sigma_{\text{condition}} (T_1 \times T_2 \times \cdots \times T_n) \right)$$

```
SELECT <column list>
FROM <T1, T2, ..., Tn>
WHERE <condition>
```

$$\prod_{\text{name,salary}} \left(\sigma_{\text{salary}<2000} (Empleados) \right)$$

```
SELECT nombre, salario
FROM Empleados
WHERE salario < 2000;
```

$$\underset{\text{P.de}}{\Upsilon}^{\text{P.de, count(Ped.id)}} (Ped \bowtie P)$$

Algebra macros reference

Write display formulas with `$$ ... $$` and use:

- `\Proj_{...}` for projection
- `\Sel_{...}` for selection
- `\Ren_{...}` for renaming
- `\Group^{...}_{...}` for grouping and aggregation
- `\NatJoin`, `\JoinBy{...}` for joins
- `\Union`, `\Inter`, `\Diff` for set operators

Algebra macros source

```
\Proj_{nombre,salario}(Empleados)
\Sel_{salario>2000}(Empleados)
\Ren_{Emp}(Empleados)
\Group^{count(Ped.id)}_{P.de}(Ped \NatJoin P)
\Sel_{Emp.id=Dept.id}(Emp \JoinBy{\text{id}} Dept)
```

Algebra macros rendered

$$\prod_{\text{nombre,salario}} (Empleados)$$

$$\sigma_{\text{salario}>2000} (Empleados)$$

$$\rho_{\text{Emp}} (Empleados)$$

$$\Upsilon^{\text{count}(Ped.id)}_{\text{P.de}} (Ped \bowtie P)$$

$$Emp \cup Dept, \quad Emp \cap Dept, \quad Emp \setminus Dept$$

Special characters and icons

Arrow examples: ➡️ ⬅️

Icon examples: 🎯🍽️🍴🧾

Algebra symbols: π σ ρ ≥ γ ∧ ∨

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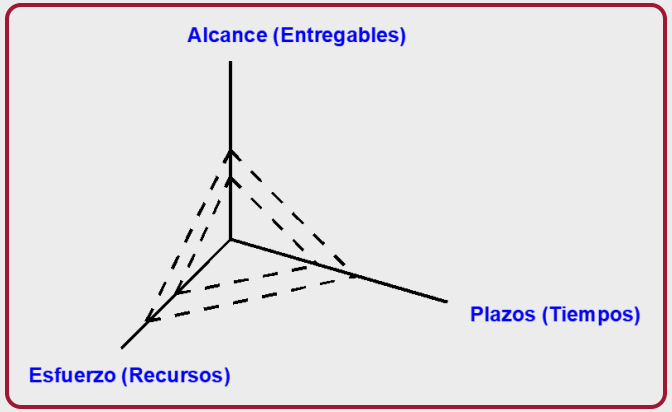
Visual regression coverage A-J

These slides are intended as regression probes.

They cover:

- image in left, right, or both columns
- text above and below framed images
- `two-col`, `two-col-3-7`, and `two-col-7-3`
- mixed cases with code and images
- image sizing and clipping behavior inside columns

Test A: left image only

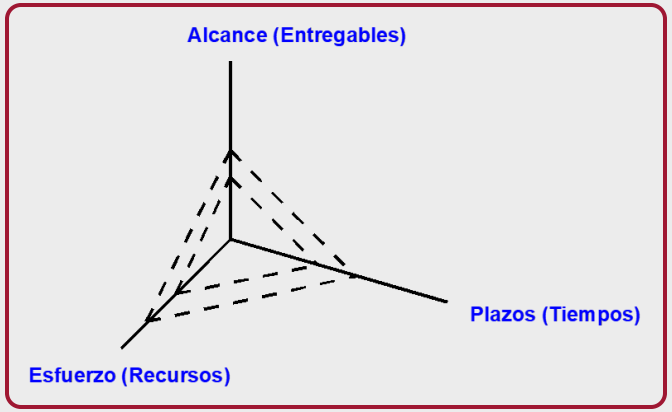


Regular text in the right column to force height distribution.

- Line 1
- Line 2
- Line 3

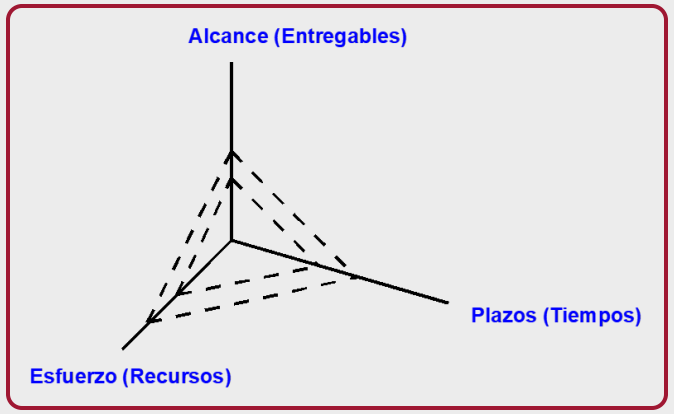
Test B: left image with text above

Text above the image.



Control content in the right column.

Test C: left image with text below

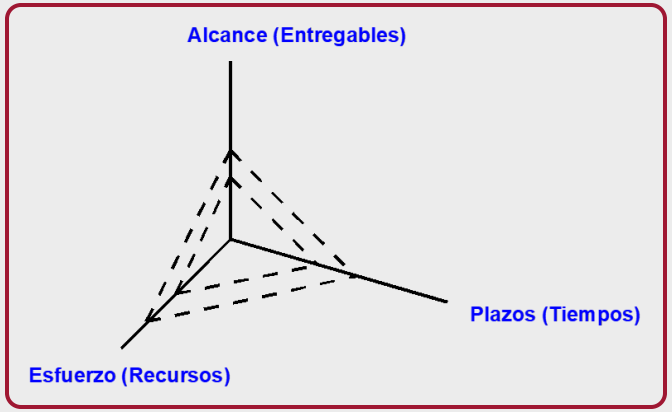


Text below the image.

Control content in the right column.

Test D: left image with text above and below

Text above the image.



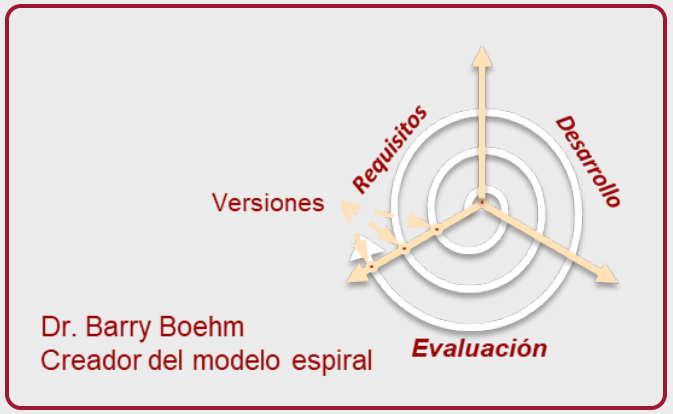
Text below the image.

Control content in the right column.

Test E: right image only

Regular text in the left column to force height distribution.

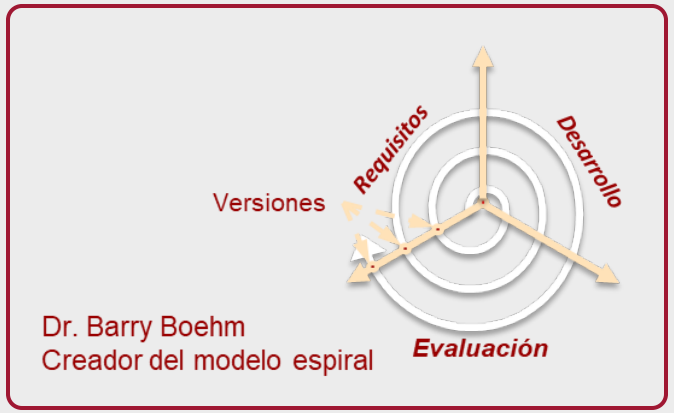
- Line 1
- Line 2
- Line 3
- Line 4



Test F: right image with text above and below

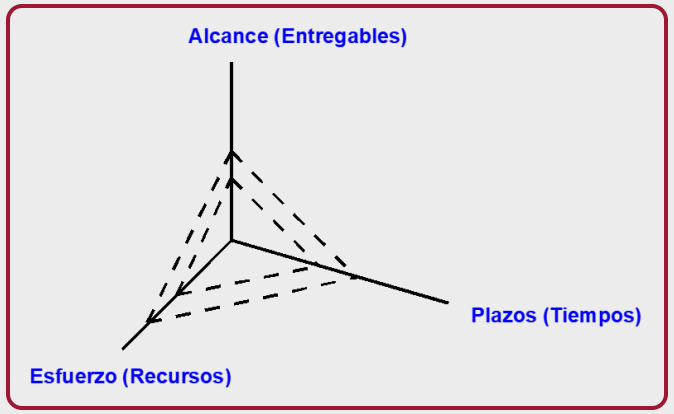
Text on the left.

Text above the image.



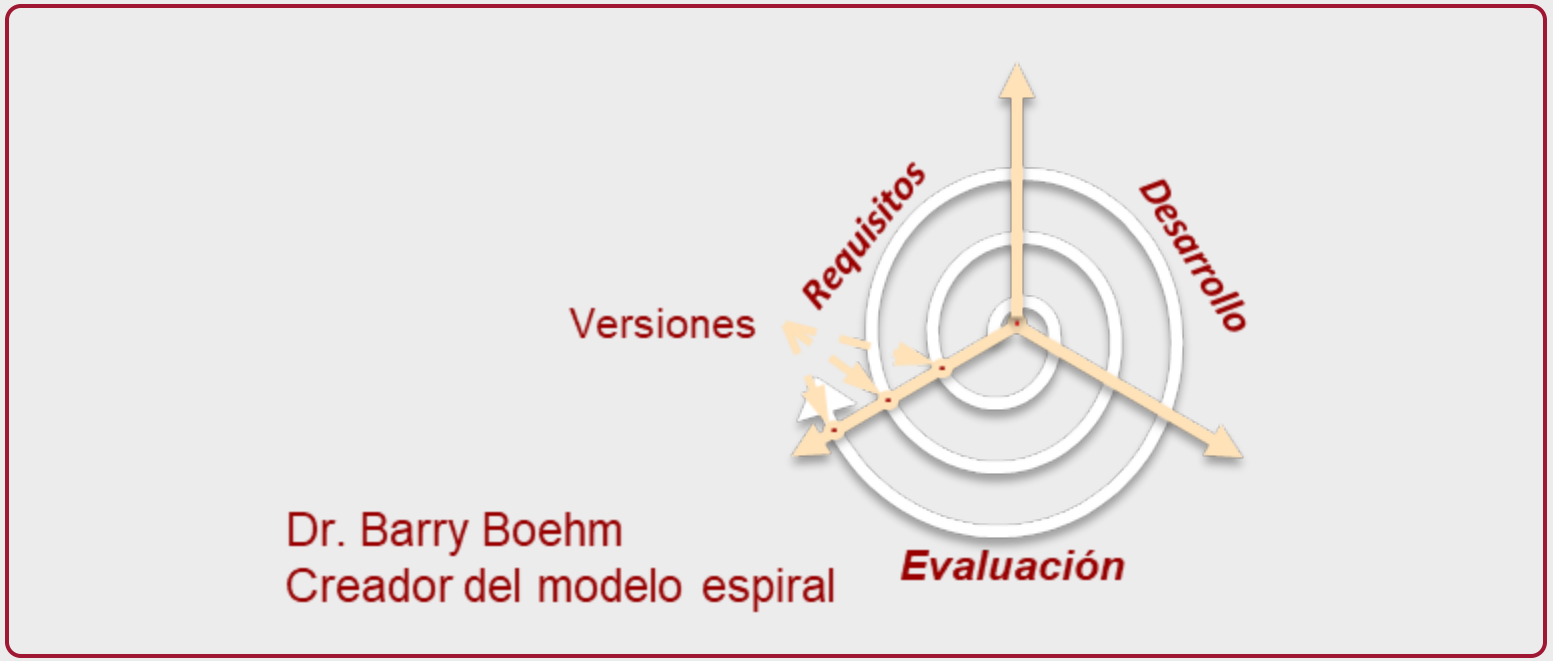
Text below the image.

Test G: image in both columns



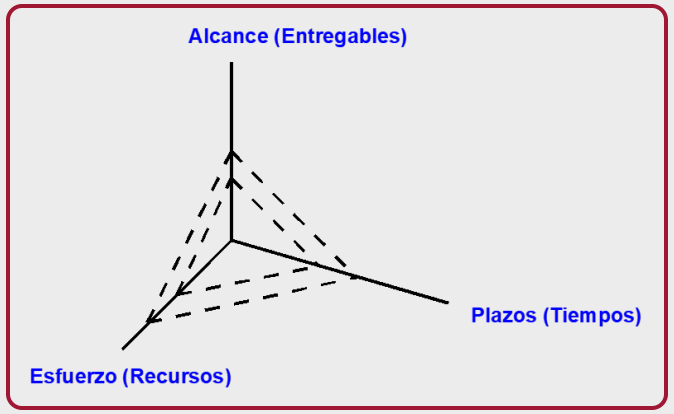
Text below the left image.

Text above the right image.



Test H: image in both columns, inverted

Text above the left image.



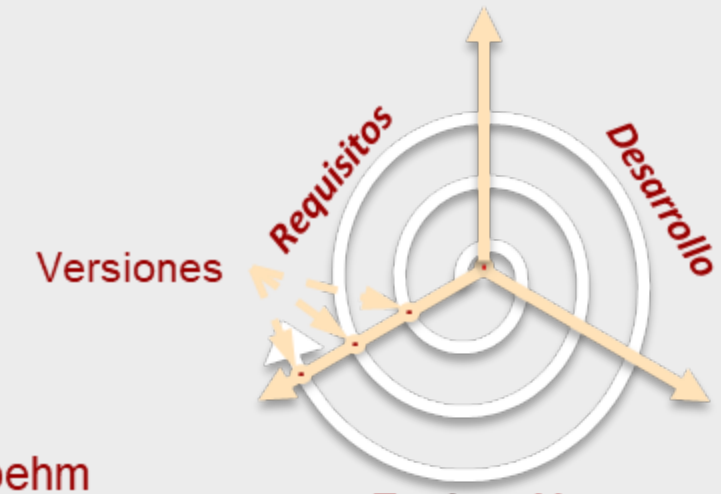
Text below the right image.

Test I: 50/50 with two images and text

Text above the left image.

Text below the left image.

Text above the right image.



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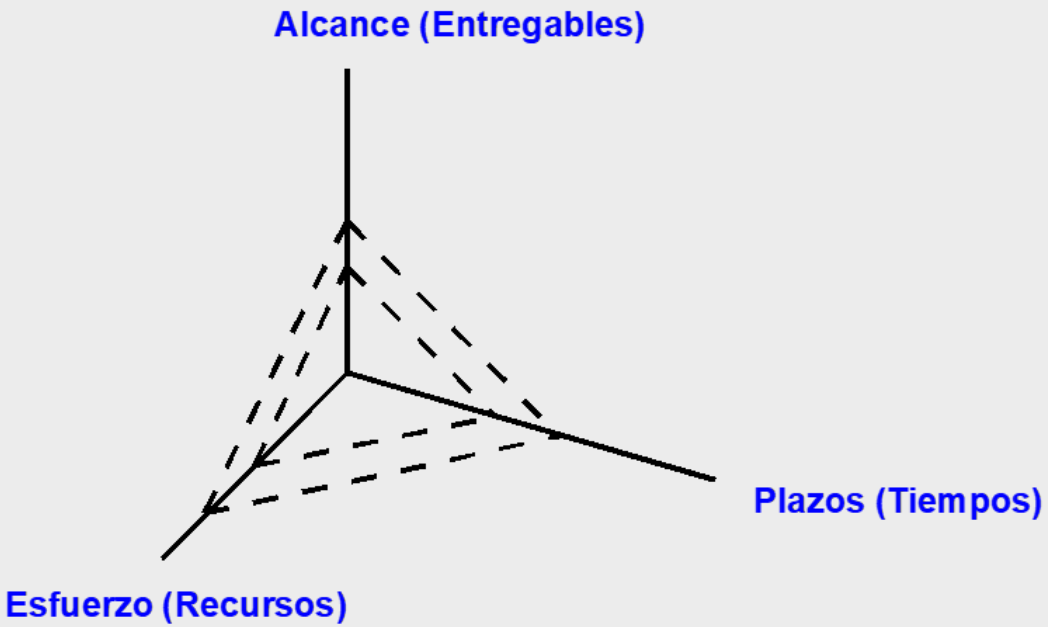
Text below the right image.

Test J: 50/50 with two images and text, inverted

Text above the left image.

Text below the left image.

Text above the right image.



Text below the right image.

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Realistic topic examples T1-T12

These slides are not only usage examples.

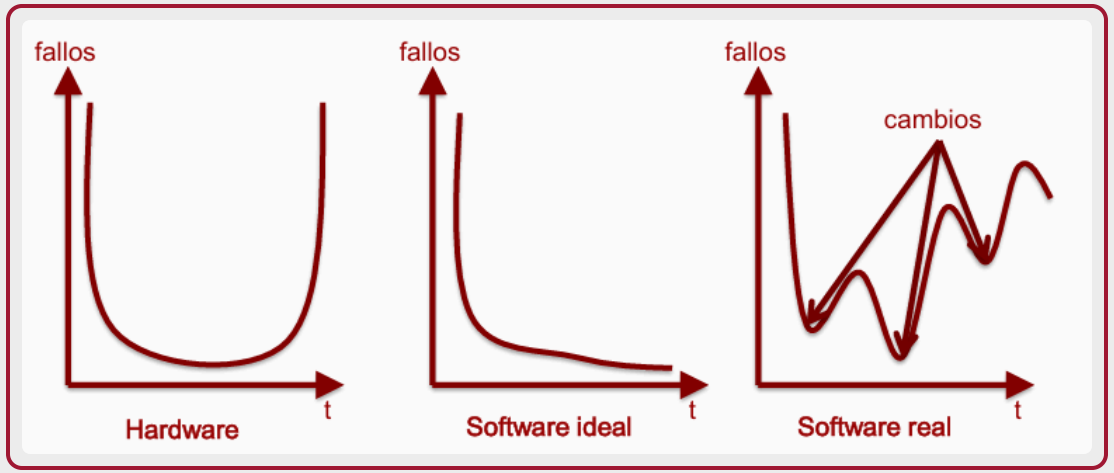
They also validate that the theme remains usable with realistic content taken from the course topics:

- image-driven slides
- asymmetric columns
- `mr` and `sql` blocks
- formulas and diagrams

T1: software fundamentals

The software crisis appears when:

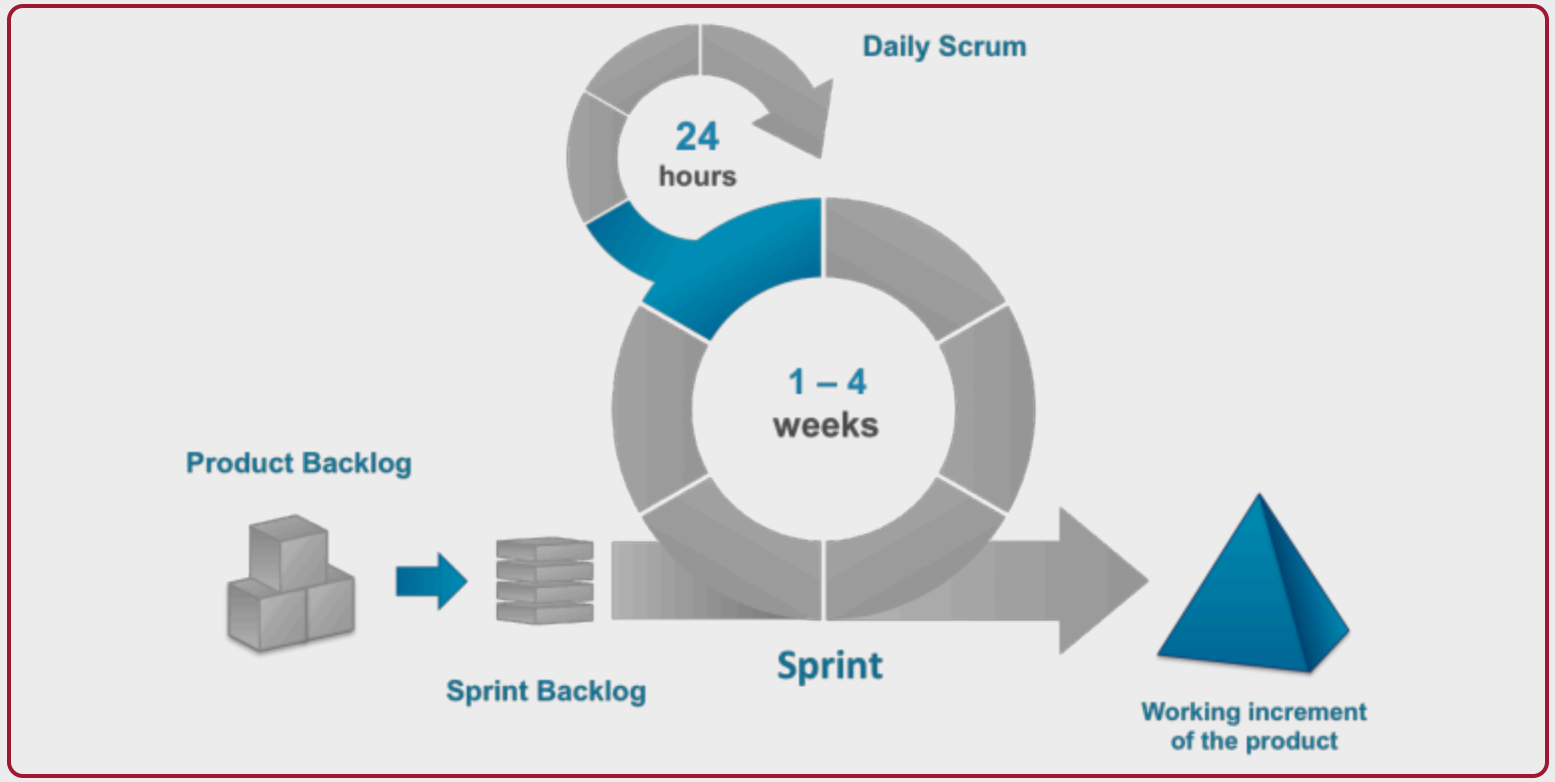
- projects become larger,
- technical complexity grows,
- engineering processes remain unclear.



T2: Scrum overview

Main roles and artifacts:

- `Product Owner`
- `Scrum Master`
- `Sprint Backlog`

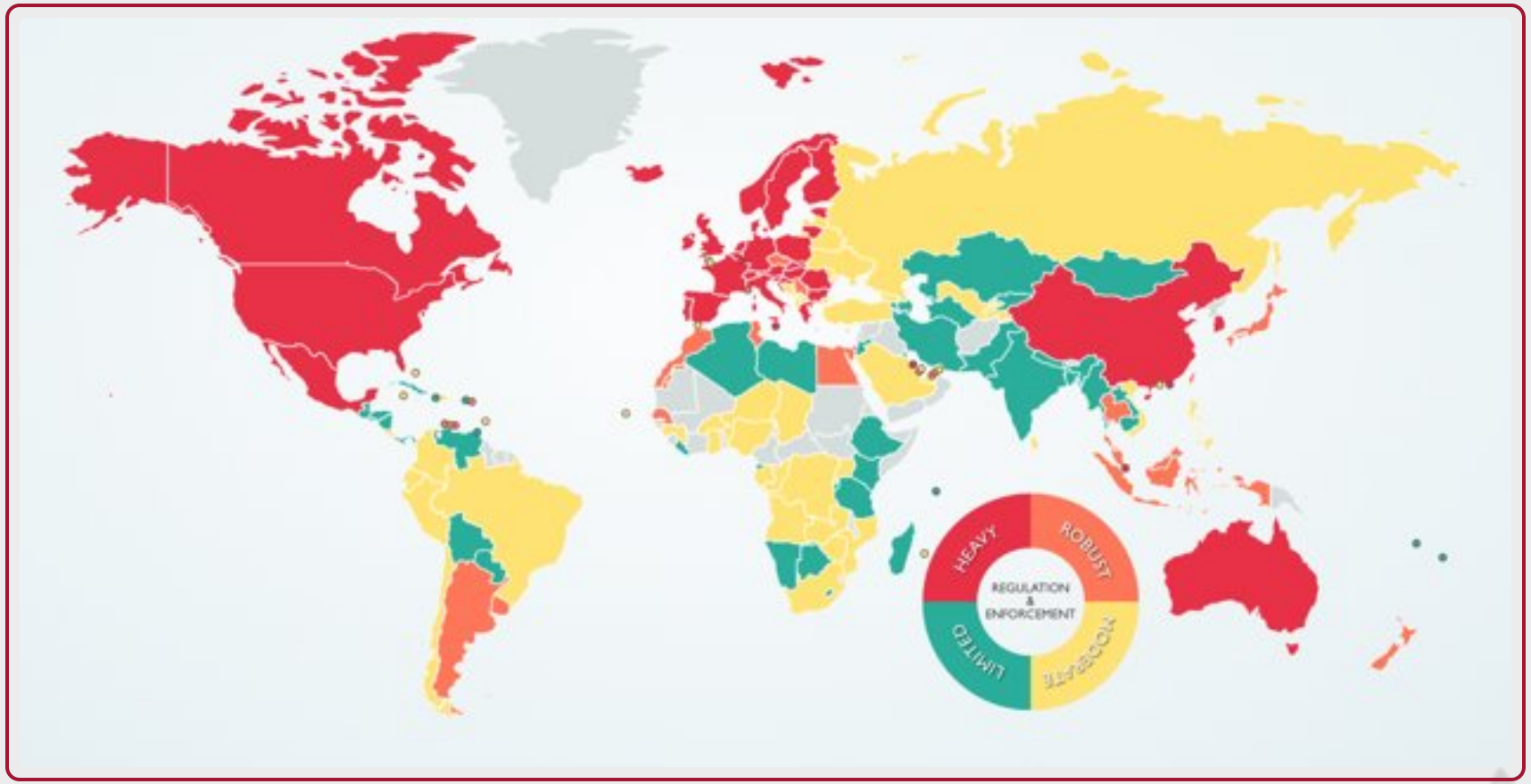


T3: information system components

An information system combines people, processes, data, and technology.

T4: data protection

Global overview of data protection regulation.



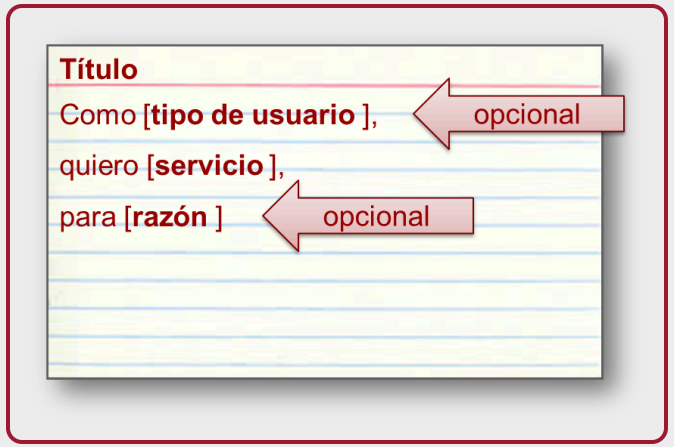
T5: user stories

Base structure:

> As a `<role>`, I want `<goal>` so that `<benefit>`.

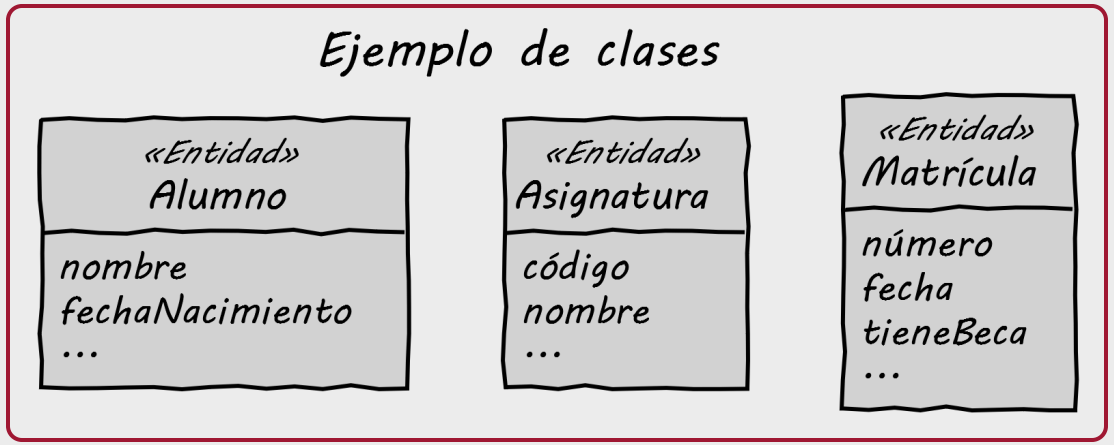
Acceptance criteria should be:

- measurable,
- verifiable,
- value-oriented.



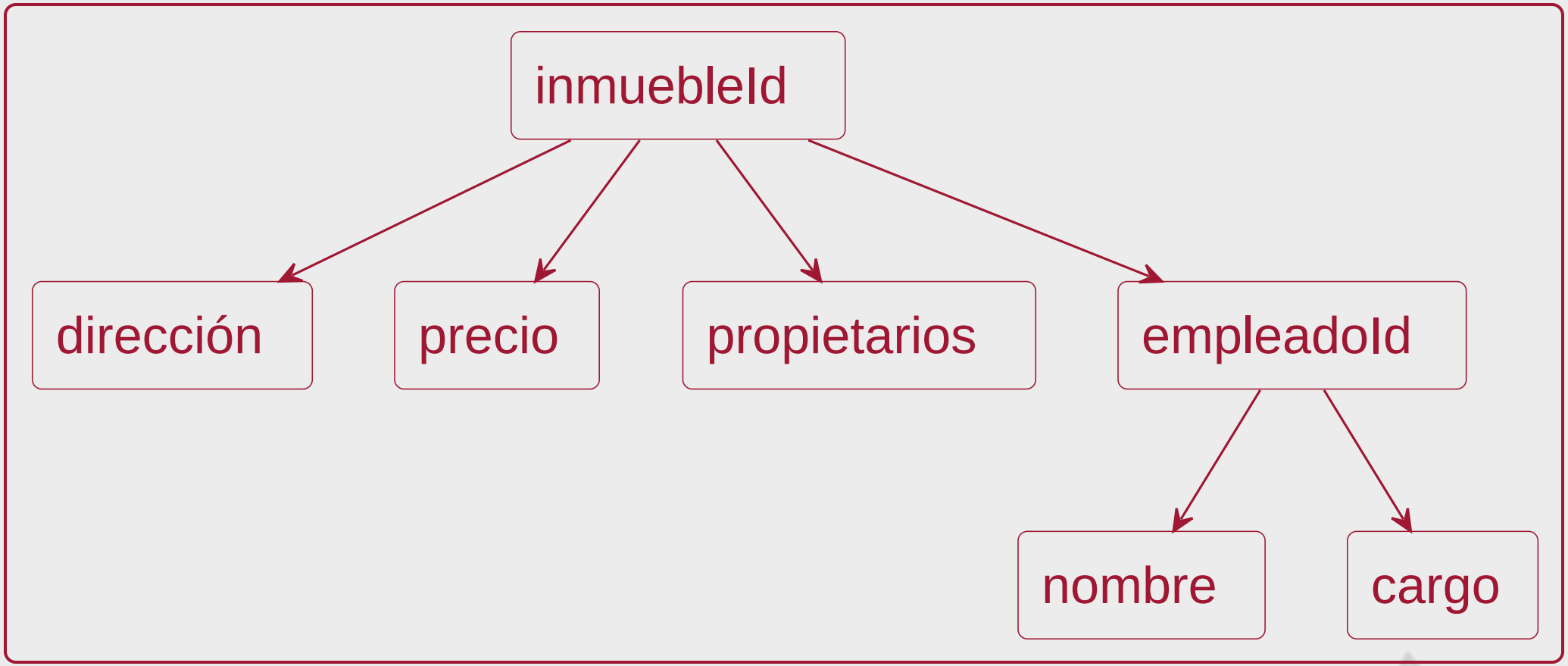
T6: UML conceptual model

Class, attribute, and association notation in UML.



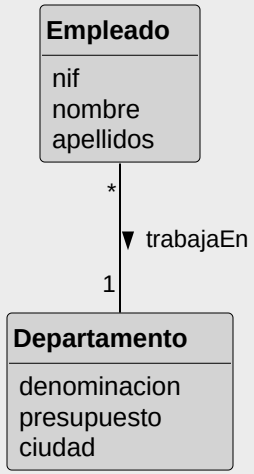
T7: functional dependencies

Dependency graph used to reason about candidate keys and normalization.



T8: conceptual-to-relational mapping

```
Departamento(id_dep, nombre)
Empleado(id_emp, nombre, FK id_dep)
```



T9: relational algebra join

$$\Pi_{\text{nombre,salario}}\left(Empleado \bowtie_{\text{Empleado.id_dept=Departamento.id_dept}} Departamento\right)$$

A	B
a_1	b_1
a_2	b_1
a_3	b_2

R

$\bowtie$

B	C
b_1	c_1
b_2	c_2
b_3	c_3

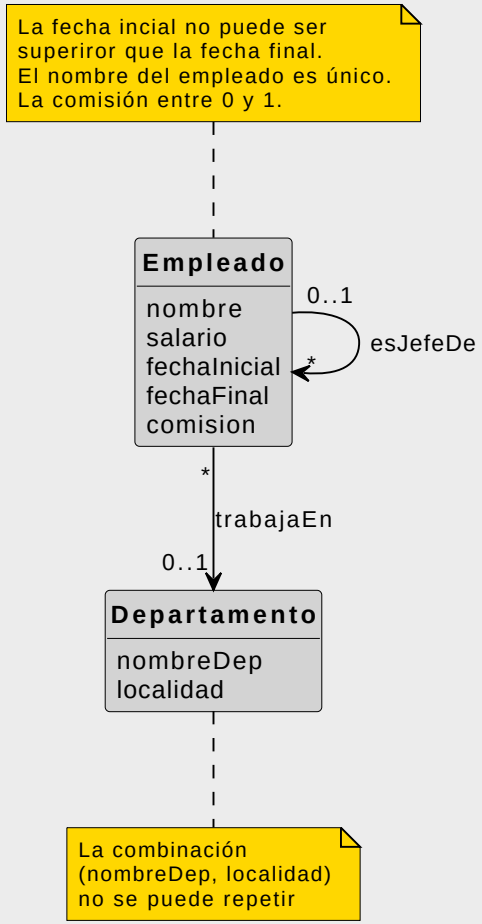
S

$=$

a_1	b_1	c_1
a_2	b_1	c_1
a_3	b_2	c_2

T10: SQL DDL and model

```sql
CREATE TABLE departamento (
  id_dept INT PRIMARY KEY,
  nombre  VARCHAR(40) NOT NULL
);
```



T11: stored procedures

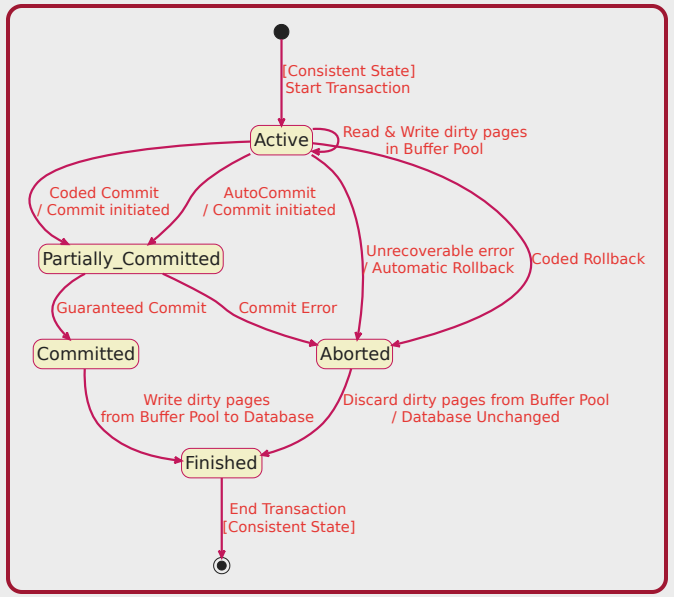
```
DELIMITER //
CREATE PROCEDURE p_raise_fee(IN p_pct DECIMAL(4,2))
BEGIN
  UPDATE empleado
  SET fee = fee * (1 + p_pct / 100);
END//
DELIMITER ;
```

```
CREATE
    [OR REPLACE]
    [DEFINER = { user | CURRENT_USER | role | CURRENT_ROLE }]
    PROCEDURE sp_name ([proc_parameter[,...]])
    [characteristic ...] routine_body

proc_parameter:
    [ IN | OUT | INOUT ] param_name type

type:
    Any valid MariaDB data type

characteristic:
    LANGUAGE SQL
  | [NOT] DETERMINISTIC
  | { CONTAINS SQL | NO SQL | READS SQL DATA | MODIFIES SQL DATA }
  | SQL SECURITY { DEFINER | INVOKER }
  | COMMENT 'string'

routine_body:
    Valid SQL procedure statement
```

T12: transaction states

Typical DBMS states:

- `Active`
- `Partially_Committed`
- `Committed`
- `Aborted`
- `Finished`



Guide map

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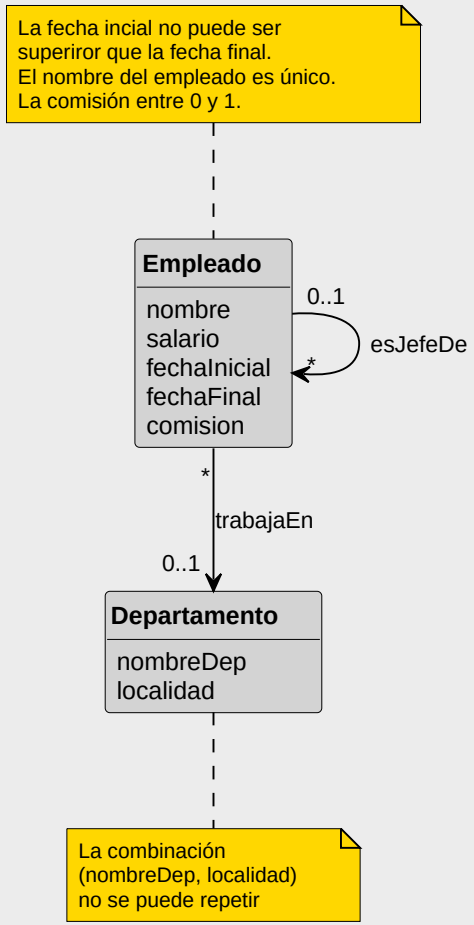
Test K: left image with right code




```
Departamentos = {departamentoId, nombreDep, localidad}
  PK(departamentoId)
  AK(nombreDep, localidad)

Empleados = {empleadoId, departamentoId, jefeId,
  nombre, salario, fechaInicial, fechaFinal, comision}
  PK(empleadoId)
  FK(departamentoId)/Departamentos
  FK(jefeId)/Empleados
```

Additional text below the code block.

Test L: left code with right image

Text above the code.

```
CREATE TABLE cuentas (
  numcta SMALLINT KEY,
  titular VARCHAR(20) NOT NULL,
  saldo DECIMAL(9,2) NOT NULL,
  CHECK (saldo >= 0)
);
```

Text below the code.

Test M: 50/50 image and code

Text above the image.

Text below the image.

Text above the code.

```
Pedido(id_ped, fecha, FK id_cli)
Cliente(id_cli, nombre, ciudad)
```

Text below the code.

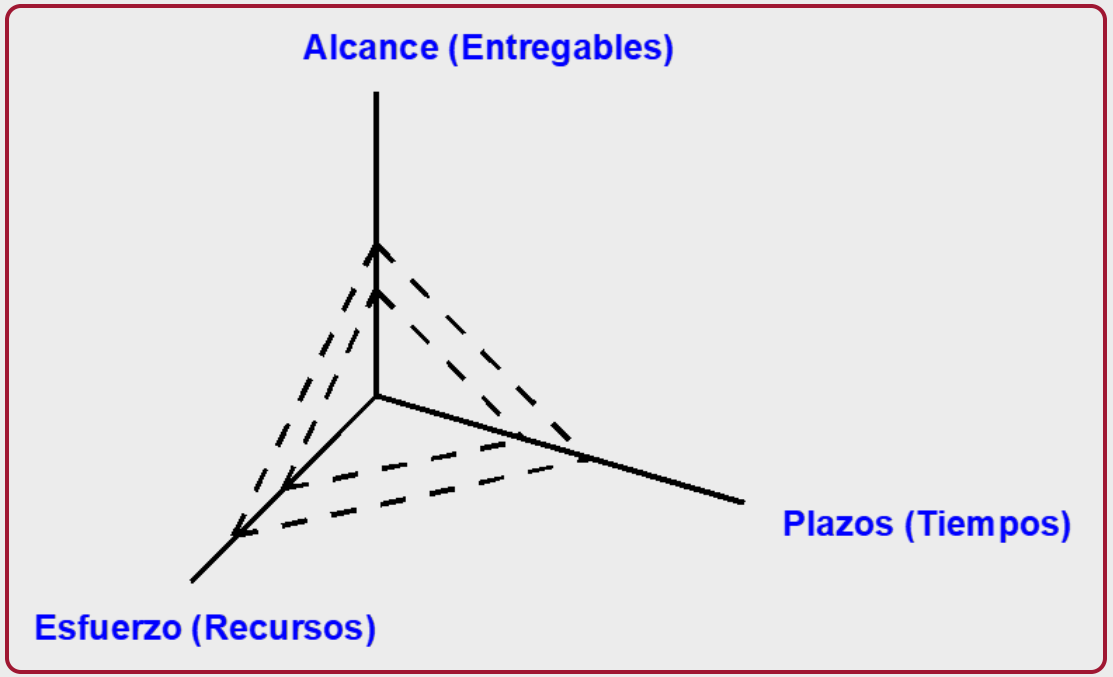
Test N: 50/50 code and image, inverted

Text above the code.

```
SELECT e.nombre, d.nombre
FROM Empleado e
JOIN Departamento d ON d.id_dept = e.id_dept;
```

Text below the code.

Text above the image.



Text below the image.